

UNIT V – Graph Theory and Combinatorics

Graphs: Definition and terminology, Representation of graphs, Multigraphs, Bipartite graphs, Planar graphs, Isomorphism and Homeomorphism of graphs, Euler and Hamiltonian paths, Graph coloring Combinatorics: Introduction, Counting Techniques, Pigeonhole Principle

◇ 1. Graphs: Definition and Terminology

◇ Definition

A **graph** is a collection of **vertices (nodes)** and **edges (links)** that connect pairs of vertices.

👉 Formally:

A graph is represented as $G = (V, E)$

- $V \rightarrow$ set of vertices
- $E \rightarrow$ set of edges connecting vertices

Example:

$V = \{A, B, C, D\}$

$E = \{AB, AC, BD, CD\}$

◇ Basic Terms

Term	Meaning	Example
Vertex (Node)	Point representing an object	A, B, C
Edge	Connection between two vertices	(A, B)
Degree	Number of edges incident on a vertex	$\deg(A) = 2$
Loop	Edge starting and ending at the same vertex	(A, A)
Parallel Edges	Two or more edges connecting same pair of vertices	Two AB edges
Path	Sequence of connected vertices	$A \rightarrow B \rightarrow C$
Cycle (Circuit)	Closed path (start = end)	$A \rightarrow B \rightarrow C \rightarrow A$
Connected Graph	Every vertex connected by some path	All vertices reachable

Disconnected Graph	Not all vertices connected	Two separate parts
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◇ 2. Representation of Graphs

Graphs can be represented in three main ways:

Type	Description	Example
Adjacency Matrix	Matrix showing 1 if edge exists, 0 otherwise.	For 4 vertices, 4×4 matrix.
Adjacency List	Each vertex lists its connected vertices.	A: B, C; B: A, D
Incidence Matrix	Rows → vertices, Columns → edges. 1 if edge touches vertex.	Used in algorithms.

👉 Example (Adjacency Matrix)

For Graph: A–B, A–C, B–C

	A	B	C
A	0	1	1
B	1	0	1
C	1	1	0

◇ 3. Types of Graphs

Type	Definition	Example / Note
Simple Graph	No loops, no parallel edges.	Commonly used.
Multigraph	Has multiple edges between same vertices.	Two roads between cities.
Directed Graph (Digraph)	Edges have direction (→).	A → B (one-way).
Weighted Graph	Edges have weights (cost/distance).	Used in Dijkstra's algorithm.
Null Graph	Only vertices, no edges.	$\deg(v)=0$ for all v .
Complete Graph (K_n)	Every vertex connected to every other.	$n(n-1)/2$ edges.
Bipartite Graph	Vertices divided into 2 sets; edges only between sets.	Students–Courses relation.
Planar Graph	Can be drawn without crossing edges.	Cube graph.

◇ 4. Bipartite Graph

A **bipartite graph** has:

- Two disjoint vertex sets **U** and **V**

- Edges only connect vertices from **U** to **V**, never within same set.

👉 Example:

$U = \{\text{Students}\}$, $V = \{\text{Courses}\}$

Edges $\rightarrow$ (Student, Course)

◇ 5. Planar Graph

A **planar graph** can be drawn on a plane such that **no two edges intersect**.

Example:

- Triangle or square is planar.
- K_5 (complete graph with 5 vertices) is **non-planar**.

🧠 **Euler's Formula for Planar Graphs:**

$$V - E + F = 2$$

where,

V = vertices, E = edges, F = faces (including outer region)

◇ 6. Isomorphism and Homeomorphism

Concept	Meaning	Example
Graph Isomorphism	Two graphs are isomorphic if they have same structure (same connectivity, different drawing).	Graph A and Graph B with same degree sequence.
Homeomorphism	Two graphs are homeomorphic if one can be obtained from another by adding or removing vertices of degree 2 (subdividing edges).	Stretching or shrinking edges without changing connectivity.

◇ 7. Euler and Hamiltonian Paths

Concept	Definition	Condition	Example
Euler Path	A path that visits every edge exactly once .	Exists if 0 or 2 vertices have odd degree.	Postman route.
Euler Circuit	Path that starts and ends at same vertex covering all edges once.	All vertices have even degree.	K_3 is Eulerian.
Hamiltonian Path	Visits every vertex exactly once .	May or may not cover all edges.	Traveling Salesman.

Hamiltonian Circuit	Hamiltonian path that ends where it started.	Exists in complete graphs.	K_4 has Hamiltonian cycle.
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◇ 8. Graph Coloring

Graph Coloring means assigning **colors to vertices** so that **no two adjacent vertices have the same color**.

- **Chromatic Number ($\chi(G)$)** → minimum number of colors required.

👉 Example:

- Triangle graph → $\chi(G)=3$
- Square (cycle of 4) → $\chi(G)=2$

Uses:

- Map coloring
- Scheduling problems
- Register allocation in compilers

◇ 9. Combinatorics: Introduction

Combinatorics is the study of **counting**, **arrangement**, and **selection** of objects.

Used in:

- Probability
- Graph theory
- Algorithm complexity

◇ 10. Counting Techniques

◇ (i) Rule of Sum (Addition Principle)

If a task can be done in **m ways** or **n ways**, then total ways = **m + n**

👉 Example: 2 red pens or 3 blue pens → total 5 choices.

◇ (ii) Rule of Product (Multiplication Principle)

If task 1 can be done in **m ways** and task 2 in **n ways**, total ways = **m × n**

👉 Example: 3 shirts × 2 pants = 6 combinations.

◇ (iii) Permutations

Order matters → arrangement of objects.

Formula:

$$^nP_r = \frac{n!}{(n-r)!}$$

👉 Example: Number of 3-digit numbers using digits 1–5 = $5P_3 = 60$

◇ (iv) Combinations

Order does **not** matter → selection of objects.

Formula:

$$^nC_r = \frac{n!}{r!(n-r)!}$$

👉 Example: Choosing 2 students from 5 = $5C_2 = 10$

◇ 11. Pigeonhole Principle

Statement:

If $n+1$ or more objects are placed into n boxes, then **at least one box** contains **2 or more objects**.

👉 **Example:**

- 13 people → 12 months.

At least **2 people have birthdays in same month**.

👉 **Use in Computer Science:**

- Hashing collisions
- Scheduling problems
- Storage allocation

12. Important Difference Tables

Basis	Euler Path	Hamiltonian Path
Covers	Every edge once	Every vertex once
Repetition	Vertices may repeat	Edges may repeat
Condition	0 or 2 vertices of odd degree	Each vertex degree ≥ 2
Example	Postman route	Traveling salesman

Basis	Graph Coloring	Bipartite Graph
Definition	Coloring vertices so no adjacent share same color	Vertices split into 2 sets (U, V)
Property	$\chi(G) \geq 2$ for connected graphs	Always 2-colorable
Example	Triangle → 3 colors	Square → 2 sets

Basis	Isomorphism	Homeomorphism
Definition	Same structure (same edges/vertices)	Can be transformed by subdividing edges
Vertex Count	Same	May differ
Example	Same network drawn differently	Stretching an edge into 2 edges

Basis	Planar Graph	Non-Planar Graph
Definition	Can be drawn without edge crossing	Some edges must cross
Example	Square	K_5 or $K_{3,3}$
Test	Euler's formula holds	Euler's formula fails

Basis	Permutation	Combination
Order	Matters	Doesn't matter
Formula	$nPr = n! / (n-r)!$	$nCr = n! / [r!(n-r)!]$
Example	ABC, ACB (different)	$\{A, B\} = \{B, A\}$ (same)

Basis	Addition Rule	Multiplication Rule
Meaning	Either this or that	This and that
Use Case	Choice between tasks	Sequence of tasks
Formula	$m + n$	$m \times n$

✔ Unit V Summary

Topic	Key Idea
Graph	Vertices + edges
Planar	No edge crossing
Bipartite	2 disjoint sets
Euler Path	Covers all edges once
Hamiltonian Path	Covers all vertices once
Coloring	Minimum colors for adjacent vertices
Combinatorics	Counting and arrangements
Pigeonhole	$n+1$ objects in n boxes $\Rightarrow$ repetition